Literature Status for arXiv:0705.0411: Finite-Space Maximal Negative Type

Run fa_banach_001, agent_lane_01

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Status

This is a `literature_already_answered` packet. It records that the finite-metric-space endpoint question raised by Doust–Weston, *Enhanced negative type for finite metric trees*, arXiv:0705.0411v2, was answered by Li–Weston, *Strict p-negative type of a metric space*, arXiv:0901.0695v3.

Source Question

In the Introduction and Synopsis of arXiv:0705.0411v2, page 3, Doust–Weston explain that their infinite necklace tree (Y, d) has strict 1-negative type but does not have p-negative type for any $p > 1$. Thus, for general metric spaces, the maximal p-negative type can be strict. They then ask:

Can the maximal p-negative type of a finite metric space be strict?

Supporting Answer

Li–Weston, arXiv:0901.0695v3, explicitly builds on the enhanced negative type method of Doust–Weston. Its abstract states that one of the paper's main results is that the supremal p-negative type of a finite metric space cannot be strict.

The exact answer appears in Section 4, titled "Supremal p-negative type of a finite metric space cannot be strict". The paper first proves Theorem 4.1: for a finite metric space (X, d), strict p-negative type is equivalent to positivity of the p-negative type gap Γ_X^p. Corollary 4.2 then shows that strict p-negative type at p persists to an interval $[p, p + \zeta)$. Consequently Corollary 4.3, on page 9, states:

The supremal p-negative type of a finite metric space cannot be strict.

This gives a negative answer to the source question: no finite metric space has a strict endpoint for its maximal/supremal p-negative type.

Scope

This is not a new proof produced by the run. It records an already-known literature answer to the finite-metric-space strict-endpoint question in arXiv:0705.0411v2. The answer is exact for finite metric spaces, hence also covers finite metric trees as a subclass. It does not address unrelated open problems about metric type, cotype, generalized roundness in other classes, or non-commutative L_p-space analogues.

Search Evidence

The lane-1 queue selected arXiv:0705.0411. Cheap run indexes were searched using the arXiv id and the core phrases “enhanced negative type”, “finite metric trees”, “negative type”, “metric trees”, and “finite trees”; no duplicate packet was found. A local corpus search for the core phrase “maximal p-negative type” identified arXiv:0901.0695v3. The supporting paper explicitly cites Doust–Weston, identifies its refinement of enhanced negative type, and states the finite-space non-strict endpoint result in its abstract, Section 4, and Corollary 4.3.

References

[1] I. Doust and A. Weston, *Enhanced negative type for finite metric trees*, arXiv:0705.0411v2.

[2] H. Li and A. Weston, *Strict p-negative type of a metric space*, arXiv:0901.0695v3.